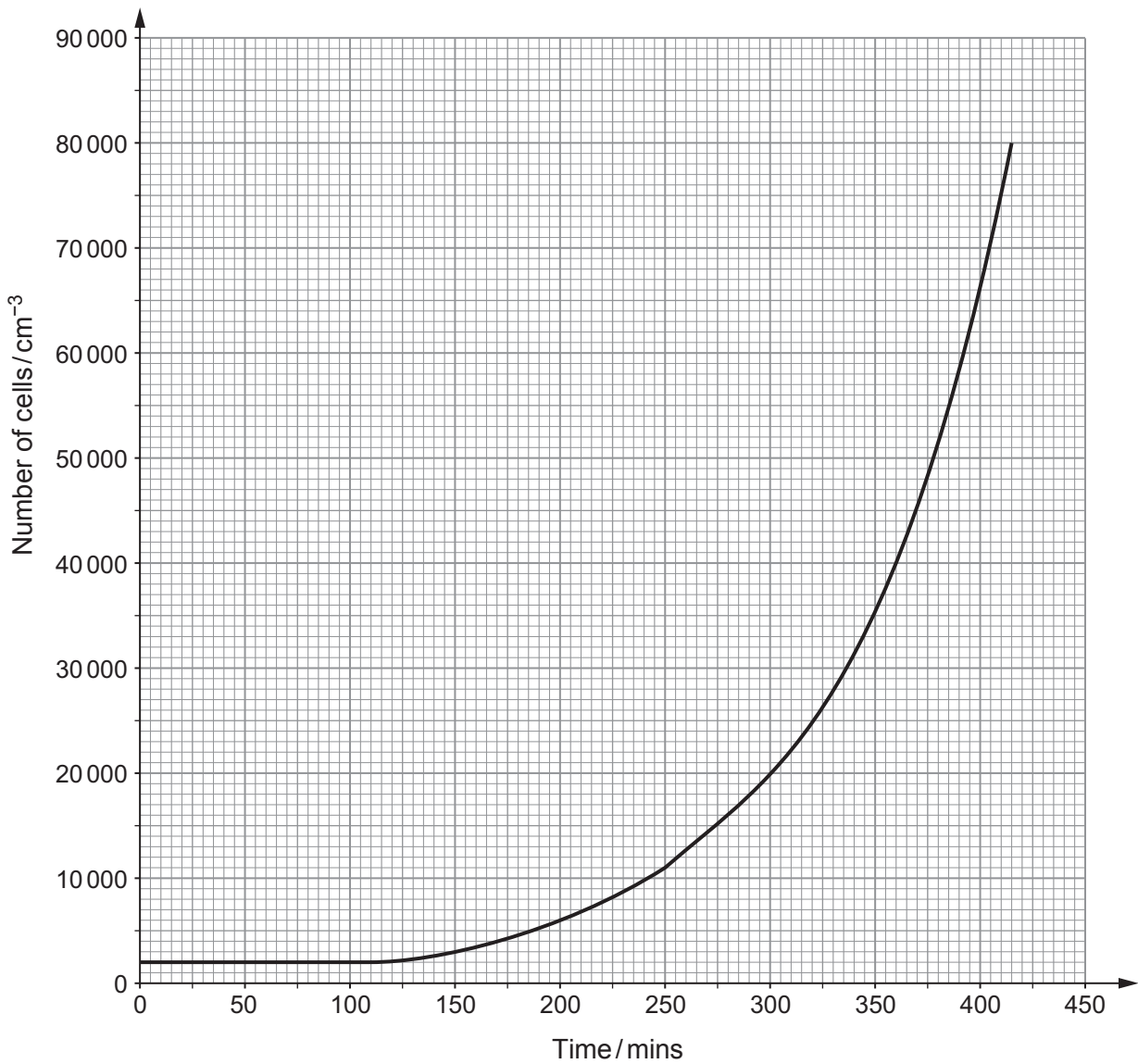


Answer **all** questions.

1. A student inoculated 900 cm^3 of fresh, sterile nutrient broth with 100 cm^3 of a culture of *Escherichia coli*. She incubated the new culture at 37°C and monitored the concentration of bacteria using a total count with an electronic cell counter. The results are shown in **Graph 1.1**.

Graph 1.1



- (a) (i) Calculate the concentration of bacteria in the **original** culture used to inoculate the broth. [2]

Concentration = cm^{-3}



- (ii) Calculate the time taken for the number of bacteria to double after the number of cells counted by the electronic cell counter had reached $20\,000\text{ cm}^{-3}$. [1]

Time taken for the number of bacteria to double = minutes

- (b) Growth curves for bacteria may be used to model growth in populations of animals, because the same phases of growth are recognisable. The following passage, taken from a website, is a description of one of the phases.

“The accumulation of waste materials, toxic metabolites and inhibitory compounds in the medium shifts the conditions of the medium such as pH and temperature. This creates an unfavourable environment for the bacterial growth. ... the number of cells undergoing division is equal to the number of cell deaths.”

- (i) Identify the phase of growth described in the passage. [1]

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- (ii) Explain why the method used to monitor the number of bacteria meant that the phase identified in part (i) was not seen in **Graph 1.1**. [1]

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- (iii) List the main steps of a different method of counting bacteria which would allow the phase described above to be seen in **Graph 1.1**. Details of aseptic technique are **not** required. [4]

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- (c) Syphilis and gonorrhoea are sexually transmitted diseases (STDs) caused by the bacteria *Treponema pallidum* and *Neisseria gonorrhoeae* respectively.

Treponema pallidum is a helically coiled micro-organism usually 6–15 μm long and 0.1–0.2 μm wide. It has a plasma membrane and an outer lipopolysaccharide membrane.

Neisseria gonorrhoeae, also known as *gonococcus*, is a species of Gram negative cocci.

Using the information given;

- (i) Describe the three-dimensional shape of *Neisseria gonorrhoeae* cells. [1]

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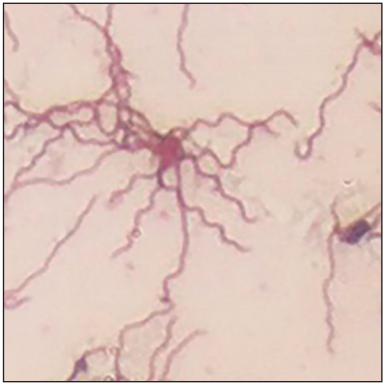
- (ii) State whether *Treponema pallidum* is Gram positive or Gram negative and explain your choice. [1]

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- (d) A 21-year-old man attended a clinic with a painless ulcer on his penis. Bacteria taken from the ulcer were Gram stained. The results are shown in **Image 1.2**.

Image 1.2



Note:
The bacteria all stained pink.

- (i) State the result of the Gram test shown in **Image 1.2** and explain why it does not help distinguish between the two types of STD described above. [2]

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- (ii) Suggest which of the two types of STD the man is most likely to be suffering from and explain your choice. [2]

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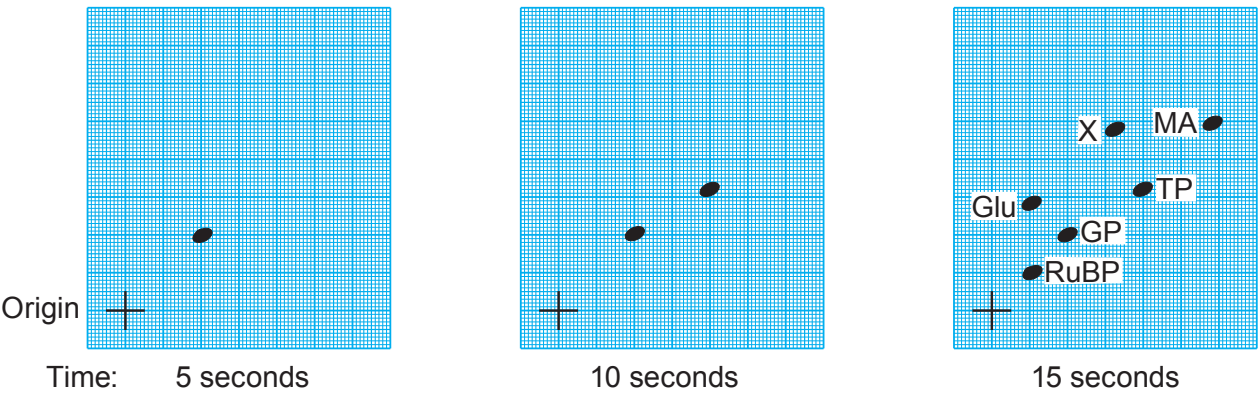
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The autoradiograms in **Image 5.2** illustrate some of Calvin's results.

Image 5.2

Experiment 1. Samples taken at short intervals from injection of $\text{NaH}^{14}\text{CO}_3$.



The times shown are the times at which the samples were dropped into hot alcohol.

Key: Glu – glucose, RuBP – ribulose bisphosphate, GP – glycerate-3-phosphate, TP – triose phosphate, MA – malic acid

(b) (i) Using evidence from **Image 5.2**, identify which of the following sequences of reactions is most likely to be correct. Explain your answer. [2]

A Ribulose Bisphosphate → Triose Phosphate → Glycerate-3-Phosphate

B Glycerate-3-Phosphate → Triose Phosphate → Ribulose Bisphosphate

C Triose Phosphate → Glycerate-3-Phosphate → Ribulose Bisphosphate

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